Environmental Justice Analysis in Alameda County

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Introduction

Summary: The location of hazardous waste treatment facilities is highly consequential due to the potential adverse effects of having such facilities located near people. Understanding where they're located and if they're located next to people is important to know who might be impacted and how to best write policy. In Alameda County (where I am from), those facilities are concentrated along the industrial corridor by the east bay (western portion of Alameda County). My hypothesis is that low income communities are located closer to hazardous waste treatment facilities because rent and home prices are likely very low in such areas.

Objective: To compare the distribution of hazardous waste proximity across Alameda County's census block groups against the distribution of its low-income population.

Loading in the necessary packages

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
```

Loading in the data we are working with

The data comes from the US EPA's EJScreen tool at the census block group level.

```
# Reading in the geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::read_sf(here::here("data", "ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNM
```

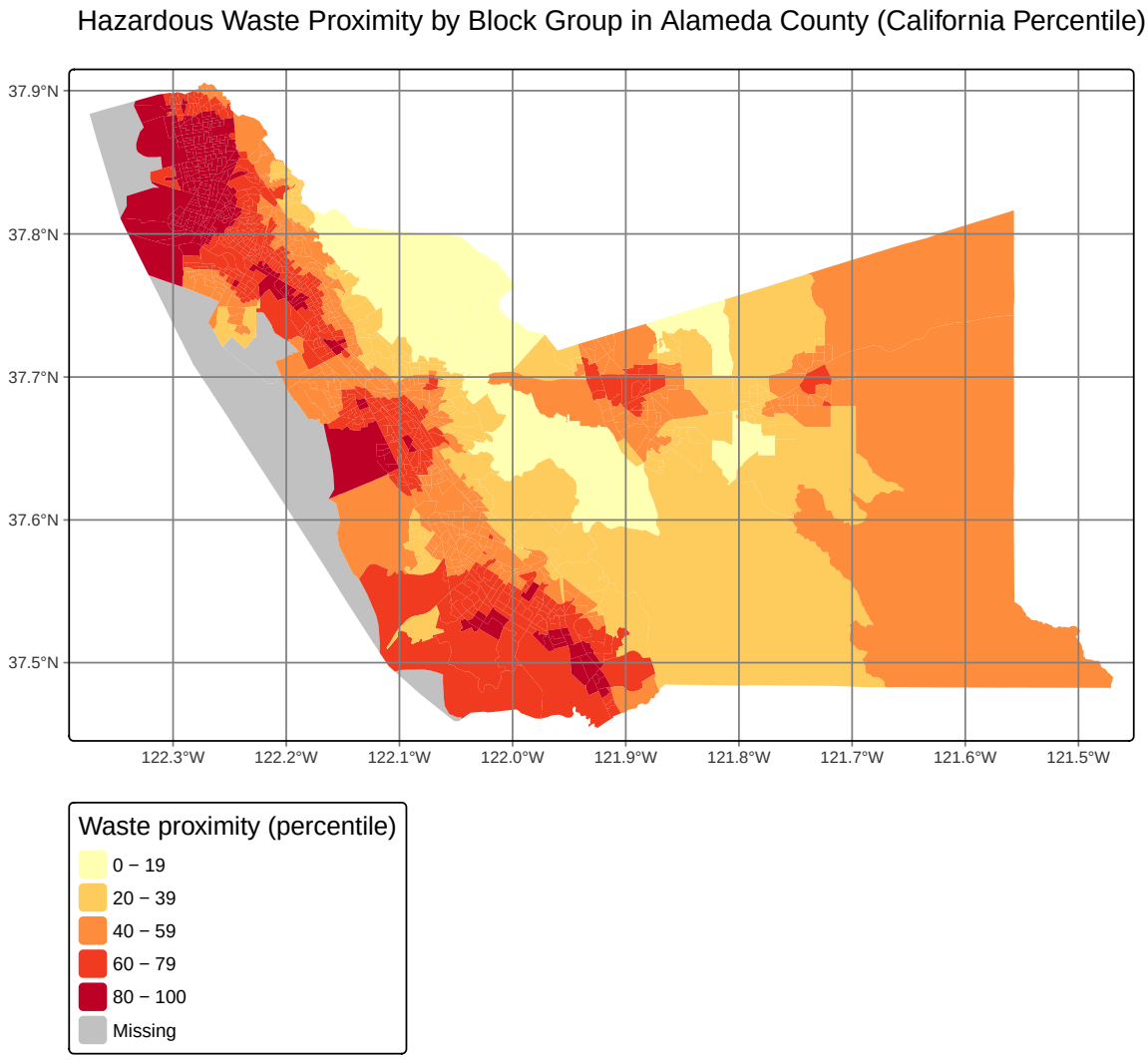
Filtering to get only Alameda County data.

```
alameda <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("Alameda County"))
```

Map 1 - Hazardous Waste Proximity in Alameda County

This map shows how close each block group is to hazardous waste facilities ranked against all block groups in California. Darker red means closer to more facilities.

```
# Mapping hazardous waste proximity as a percentile within California
tm_shape(alameda) +
  tm_title("Hazardous Waste Proximity by Block Group in Alameda County (California Percentile
  tm_fill(fill = "P_PTSDF",
          fill.scale = tm_scale(values = "brewer.yl_or_rd"),
          fill.legend = tm_legend(title = "Waste proximity (percentile)")) +
            tm_graticules()
```

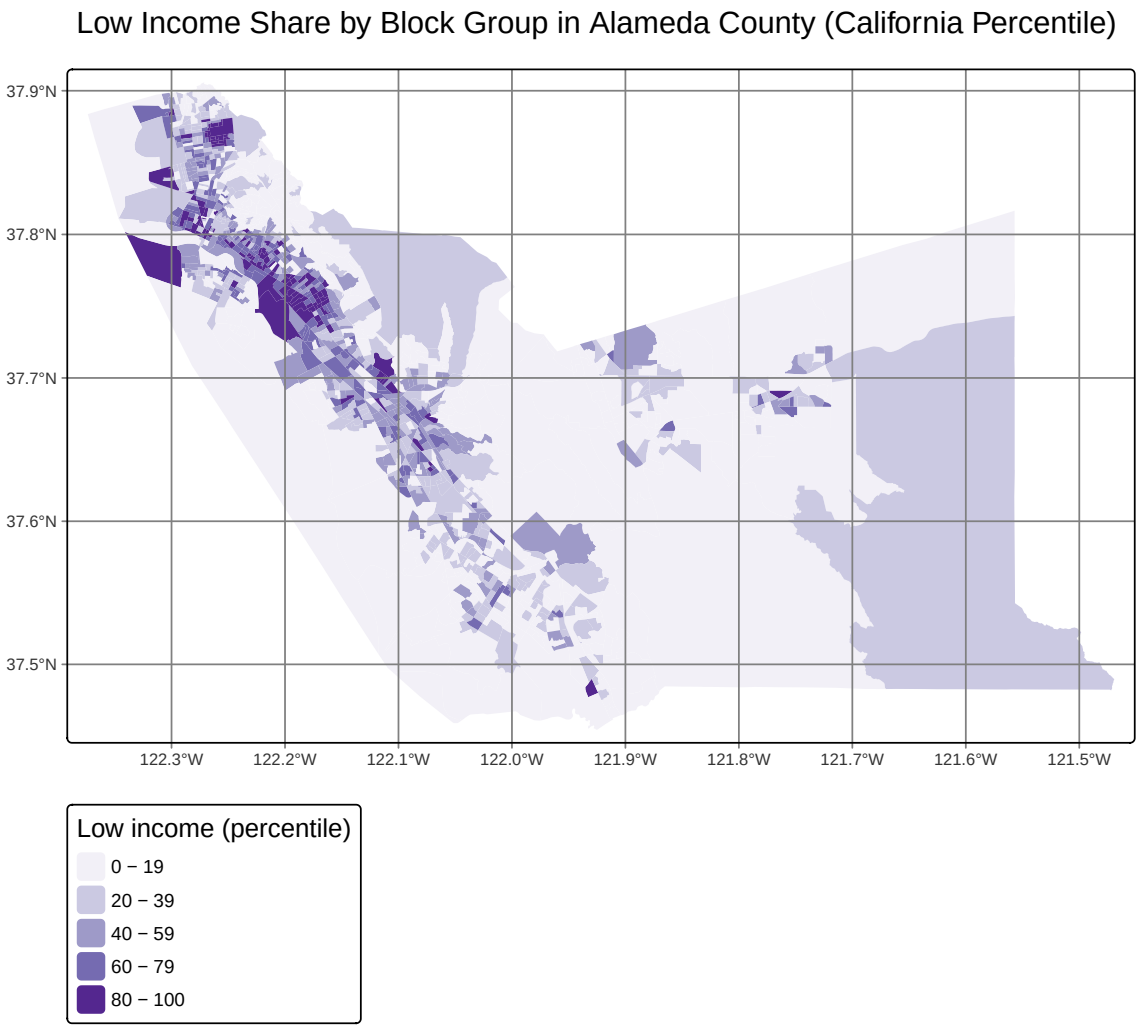


Map 2 - Low Income Residents in Alameda County

This map shows the share of low income residents in each block group ranked against all block groups in California. Darker purple means a higher share of low income residents.

```
# Mapping the low income population as a percentile within California
tm_shape(alameda) +
```

```
  tm_title("Low Income Share by Block Group in Alameda County (California Percentile)") +
  tm_fill(fill = "P_LOWINCPCT",
          fill.scale = tm_scale(values = "brewer.purples"),
          fill.legend = tm_legend(title = "Low income (percentile)")) +
  tm_graticules()
```



Analysis

I believe my hypothesis is somewhat valid because it's clear to see both maps feature a diagonal line of darker block groups located in the western portion of Alameda County. Block groups with a high proportion of low income residents match up with block groups that are in higher proximity to hazardous waste in that diagonal. However, when taking a look at other sections of Alameda County this trend doesn't perfectly continue, as not all block groups with high proximity to hazardous waste have a high proportion of low income residents and vice versa. Some correlation does not mean there is causation, so I cannot effectively conclude something, but it is clear that in certain areas lower income block groups are in higher proximity to hazardous waste facilities.

Citation

U.S. Environmental Protection Agency (EPA). 2023. EJScreen: Environmental Justice Screening and Mapping Tool, Version 2.2. The tool is now defunct and is no longer available from the EPA.